Chapter 36 — Room Migrations & Schema Management

> **Part IV — Offline-First & Synchronization** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 3 (Room/SSOT — defined migrations briefly), Ch 16 (API schema evolution — the network analogue), Ch 34 (cache versioning). The local- storage counterpart to Ch 16; critical for offline-first apps whose local DB *is* the source of truth (Ch 29).
>
> **Reference surface in StreamCart:** evolving the catalog/cart/wallet/outbox database schemas across app releases **without losing users' offline data**.

36.1 Problem Definition and Real-World Context

In an offline-first app, the local database holds the user's data and queued writes (Ch 29/31). When a new app version changes the schema (adds a column, a table, changes a type), Room must **migrate** the existing on-device database to the new schema **without data loss** — because that data may not exist anywhere else yet (unsynced writes) and refetching is expensive or impossible offline. Mishandling migrations is a top cause of crashes and data-loss incidents:

- **Crash on open.** Room throws `IllegalStateException` if the schema version increased but no `Migration` is provided — the app crashes on launch for every updating user.
- **Silent data wipe.** `fallbackToDestructiveMigration()` "fixes" the crash by **deleting the entire database** — wiping carts, drafts, downloads, and unsynced outbox entries. A catastrophic, irreversible loss shipped to production.
- **Untested migrations.** A hand-written SQL migration with a typo corrupts data or crashes — and it's only discovered in production because the test suite ran on a fresh DB, never exercising the migration path.
- **Lossy complex changes.** Changing a column type or constraint in SQLite requires a careful table-recreate-and-copy; a naive approach loses data or breaks.

Migration/schema management defines **how the local DB evolves across releases**, the **data-preservation guarantee**, the **testing discipline** (migrations are code that *must* be tested), and the **version lineage** — properties that decide whether an app update is safe for users' data.

Three practical use cases

1. **E-commerce (StreamCart).** Adding a `discount` column to products and a `wishlist` table across releases while preserving cart/wishlist and **unsynced outbox** entries.
2. **Fintech (Wallet).** Changing a column type (cents → minor-unit) via a careful table-recreate migration with data transformation, on encrypted data (Ch 24).
3. **OTT (Watch).** Adding indexes and an FTS table (Ch 35) via migration; renaming a table while keeping watch-history and downloads.

Why this is architectural

Schema management defines the **data-durability contract across app versions**: that an update never loses user data, how schema changes are expressed and **tested**, and how versions chain — properties that determine whether offline data survives the app's evolution.

36.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Migrate existing databases to new schemas without data loss.
F2	Support additive (column/table/index) and complex (type/constraint) changes.
F3	Chain migrations across multiple versions ($1 \to 2 \to 3$, and jumps $1 \to 3$).
F4	Test every migration (schema + data) before release.
F5	Never silently destroy user data (no destructive fallback in prod).
F6	Export schemas for diffing/validation and auto-migration.

Non-functional requirements

Attribute	Policy
Reliability	No data loss; no crash-on-open after update.
Correctness	Migrations tested with real DB + data; schemas validated.
Performance	Migrations fast (don't block startup for huge tables).
Maintainability	Clear version lineage; auto-migrations where possible.
Safety	Destructive fallback forbidden in release; encrypted DB migratable (Ch 24).

Scope

- **In scope:** `Migration` classes, auto-migrations, `@RenameColumn` / `@DeleteTable` specs, complex (table-recreate) migrations + data transforms, `MigrationTestHelper`, schema export, version management, destructive-fallback dangers, migrating encrypted DBs.
- **Out of scope:** the SSOT/Room basics (Ch 3), API schema evolution (Ch 16), cache versioning (Ch 34), encryption mechanics (Ch 24).

Assumptions & constraints

- Room 2.6+ (auto-migrations + spec annotations), `exportSchema = true`. The local DB may hold **unsynced** data (Ch 31) → loss is unacceptable. `minSdk 24`.

36.3 System Design Fundamentals

- **Schema version.** `@Database(version = N)`. Incrementing it signals Room to migrate; opening a DB at version `M < N` without a path from M→N throws (or destroys, if fallback is set).
- **`Migration(from, to)`.** A class whose `migrate(db)` runs raw SQL to transform the schema (and data) from `from` to `to`. Room applies the chain of migrations covering the user's installed version to the current one (e.g., a user on v1 opens v3 → runs 1→2 then 2→3, or a single 1→3).
- **Auto-migrations (Room 2.4+).** For **simple additive** changes (add column/table, with defaults), Room **generates** the migration from the exported schemas — you declare `@AutoMigration(from, to)` and, for ambiguous changes (rename/delete), an `AutoMigrationSpec` with `@RenameColumn` / `@RenameTable` / `@DeleteColumn` / `@DeleteTable`. Prefer auto-migrations for simple changes; hand-write complex ones.
- **Complex migrations (table recreate).** SQLite can't `ALTER` a column's type/ constraint. The pattern: **create a new table** with the new schema, **copy/ transform** data from the old, **drop** the old, **rename** the new — all in the `Migration`. Data transformations (e.g., splitting a column) happen here.
- **Schema export (`exportSchema = true`).** Room writes JSON schema files per version into the repo; these enable **auto-migrations**, **schema diffing in code review**, and `MigrationTestHelper` to create/validate old-version DBs. Commit these files.
- **Testing is mandatory.** `MigrationTestHelper` creates a DB at the old version (from the exported schema), inserts data, runs the migration, and **validates** the resulting schema and data. **An untested migration is a production incident waiting to happen** — the normal test suite runs on a fresh DB and never exercises migrations.
- **Never `fallbackToDestructiveMigration()` in release.** It deletes all data on a missing/failed migration. Acceptable only for purely-disposable caches or in debug; for user data (especially unsynced, Ch 31) it's catastrophic. Provide real migrations and **fail loudly in debug**.
- **Relation to Ch 16/34.** API schema evolution (Ch 16) handles the *network* DTO; Room migrations handle the *storage* schema; cache versioning (Ch 34) handles cached-shape invalidation. All three coordinate when a data model changes.

Core insight: **increment the version, provide a tested migration (auto for simple, hand-written table-recreate for complex) that preserves data, export and commit schemas, and never destructively fall back** — so every app update migrates users' offline data safely.

36.4 High-Level Architecture


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Components & ownership

- **Version + migrations** chain the schema M→N preserving data.
- **Auto-migrations + specs** handle simple/rename/delete changes.
- **Manual migrations** handle complex type/constraint changes (recreate+copy).
- **Exported schemas** (committed) enable auto-migration + testing + review.
- `MigrationTestHelper` validates each migration with data.

Direction: open → detect version gap → apply migration chain (auto/spec/manual) → data preserved; schemas exported for generation + tests.

36.5 Detailed Component Design



Change	Mechanism	Example
Add column (with default)	Auto-migration	`@AutoMigration(1,2)`
Add table / index	Auto-migration	generated
Rename column/table	Auto-migration + `AutoMigrationSpec`	`@RenameColumn`
Delete column/table	Auto-migration + spec	`@DeleteColumn`
Change type / constraint	**Manual** `Migration`	create-copy-drop-rename
Data transformation	Manual `Migration`	split/compute in SQL

36.6 End-to-End Data Flow

Migration chain on app update (data preserved)



Complex migration (table recreate + transform)



36.7 Complete Kotlin Implementation

Package: `com.streamcart.core.database`.

36.7.1 Database with version, auto-migrations, manual migrations

```
@Database(
    entities = [ProductEntity::class, CartItemEntity::class, OutboxEntry::class],
    version = 3,
    exportSchema = true,                                  // commit schemas/ JSON (F6)
    autoMigrations = [
        AutoMigration(from = 1, to = 2),                 // simple additive (add discount col)
        AutoMigration(from = 2, to = 3, spec = Migrations.RenameBrandSpec::class),  // rename
    ],
)
abstract class StreamCartDatabase : RoomDatabase() { /* DAOs */ }

// Build with manual migrations too (for complex changes not auto-migratable).
Room.databaseBuilder(context, StreamCartDatabase::class.java, "streamcart.db")
    .addMigrations(Migrations.MIGRATION_3_4_TYPE_CHANGE)   // manual
    // NO fallbackToDestructiveMigration() in release!
    .build()
```

36.7.2 Auto-migration spec (rename/delete — ambiguous changes)

```
object Migrations {
    @RenameColumn(tableName = "products", fromColumnName = "brand", toColumnName = "brand_name")
    class RenameBrandSpec : AutoMigrationSpec

    // ... manual migrations below
}
```

36.7.3 Simple manual migration (additive — when not using auto)

```
val MIGRATION_1_2 = object : Migration(1, 2) {
    override fun migrate(db: SupportSQLiteDatabase) {
        db.execSQL("ALTER TABLE products ADD COLUMN discount INTEGER NOT NULL DEFAULT 0")
    }
}
```

36.7.4 Complex manual migration (type change via recreate + transform)

```
// Change price from TEXT "12.99" to INTEGER cents (1299). SQLite can't ALTER type -> recreate.
val MIGRATION_3_4_TYPE_CHANGE = object : Migration(3, 4) {
    override fun migrate(db: SupportSQLiteDatabase) {
        db.execSQL("""
            CREATE TABLE products_new (
                id TEXT NOT NULL PRIMARY KEY,
                title TEXT NOT NULL,
                price_cents INTEGER NOT NULL DEFAULT 0,
                discount INTEGER NOT NULL DEFAULT 0
            )
        """.trimIndent())
        // transform old TEXT price -> cents during copy (data transformation)
        db.execSQL("""
            INSERT INTO products_new (id, title, price_cents, discount)
            SELECT id, title, CAST(ROUND(CAST(price AS REAL) * 100) AS INTEGER), discount
            FROM products
        """.trimIndent())
        db.execSQL("DROP TABLE products")
        db.execSQL("ALTER TABLE products_new RENAME TO products")
        db.execSQL("CREATE INDEX index_products_title ON products(title)")   // recreate indexes
    }
}
```

36.7.5 Adding an FTS table via migration (ties to Ch 35)

```
val MIGRATION_4_5_FTS = object : Migration(4, 5) {
    override fun migrate(db: SupportSQLiteDatabase) {
        db.execSQL("CREATE VIRTUAL TABLE products_fts USING fts4(content=`products`, title,
description)")
        db.execSQL("INSERT INTO products_fts(products_fts) VALUES('rebuild')")   // populate
index
        // + AFTER INSERT/UPDATE/DELETE triggers to keep it in sync (Ch 35)
    }
}
```

36.7.6 Module structure

```
:core:database   StreamCartDatabase (version/autoMigrations), Migrations, AutoMigrationSpecs
schemas/         exported JSON per version (COMMITTED to VCS)
src/androidTest  MigrationTest (MigrationTestHelper)
```

```
// build.gradle: export schema to a committed directory
// ksp { arg("room.schemaLocation", "$projectDir/schemas") }   // and add schemas/ to sourceSets
+ VCS
```

36.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Migration runs on main thread at open	ANR for big tables	open DB off main; keep migrations fast; show splash
Migration crash mid-way	partial schema	Room runs each migration in a transaction (rolled back on failure)
Skipped-version jump (1→3)	missing path	provide all pairwise or chained migrations; Room chains them
Destructive fallback	data wiped	forbidden in release; provide migrations
Unsynced outbox lost	write loss	migrate (don't drop) the outbox table; test it
Concurrent open before migration done	inconsistent	Room serializes open/migration

- **Each migration runs in a transaction** — a failing `migrate()` rolls back, so the DB isn't left half-migrated (it fails to open, surfaced in debug). Still, **test** to avoid shipping a failing migration.
- **Keep migrations fast**; a huge table-recreate can block startup — do heavy data transforms incrementally or off the critical path if needed.

36.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Missing migration (version bumped)	`IllegalStateException` on open	add the `Migration`/auto-migration	no crash
Migration SQL error	crash on open (rolled back)	fix + test with `MigrationTestHelper`	safe after fix
Destructive fallback shipped	data wiped	remove it; provide migrations; postmortem	prevent recurrence
Data loss in complex migration	QA/data check	fix copy/transform SQL; test data preservation	preserved
Skipped versions	—	Room chains migrations; provide the path	migrates
Encrypted DB migration (Ch 24)	open with key	migrations run inside the encrypted DB (SQLCipher)	works
Huge table slow migration	ANR	optimize SQL / incremental; off-main open	no ANR
Downgrade (version decreased)	rare	handle/forbid; don't destroy	defined



> **The cardinal rule: provide a migration, never destroy.** A missing migration should fail loudly in debug (so you catch it) — never be papered over with destructive fallback that wipes users' (possibly unsynced) data in production.

36.10 Scaling the Implementation

- **Level 1.** Manual `Migration` for a simple add-column; `exportSchema = true`.
- **Level 2.** Auto-migrations for additive changes + `AutoMigrationSpec` for renames/deletes; commit schemas; test with `MigrationTestHelper`.
- **Level 3 — production.** Complex table-recreate migrations with data transforms; migration tests covering schema **and** data; forbid destructive fallback; CI schema validation.
- **Level 4 — large.** Incremental/background migration for huge tables; coordinated schema changes across modules; migrating encrypted DBs (Ch 24); migration + cache version (Ch 34) + API evolution (Ch 16) coordination.
- **Level 5 — enterprise.** Schema governance/versioning policy, migration review gates, zero-downtime large-data migrations, rollback/forward-compat strategy, migration observability (success rate per version).

Avoid overengineering: simple apps use auto-migrations for additive changes. Reserve manual table-recreate for genuine type/constraint changes; don't hand-write what Room can generate.

36.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Simple change	Auto-migration	manual `Migration`	Auto for additive (less error-prone); manual when auto can't express it
Rename/delete	Auto + `AutoMigrationSpec`	manual	Spec for clarity; manual for complex
Type/constraint change	Manual (recreate+copy)	auto (can't)	Manual — SQLite can't ALTER type
Missing migration	Provide it	`fallbackToDestructiveMigration`	**Always** provide; destructive only for disposable caches/debug
Schema export	`exportSchema=true` (commit)	off	On — enables auto-migration, tests, diffs
Big-data migration	In-migration transform	background/incremental	In-migration for small; background for huge tables

Auto vs manual migration: auto-migrations (from exported schemas) are less error-prone for additive/simple changes and reduce boilerplate, but can't express type changes, complex data transforms, or ambiguous renames

without a spec. Manual migrations are fully general but must be hand-written and **tested**. Use auto for what it handles; manual for the rest.

Destructive fallback — when (rarely) OK: only when the database is a **purely disposable cache** that can be fully rebuilt from the network with no unsynced data and no cost to the user. For anything holding user data or unsynced writes (Ch 31), it's a data-loss bug. Even then, prefer cache versioning (Ch 34) to a DB wipe.

36.12 Security Considerations

- **Migrate encrypted databases** (Ch 24): migrations run **inside** the SQLCipher- opened DB — the key must be available at open; the migration SQL operates on decrypted pages. Test migrations on the encrypted DB.
- **Don't leak data during a type/transform migration** — a recreate that copies into a temp table must not leave plaintext temp data on an encrypted DB (SQLCipher keeps temp encrypted; verify).
- **Schema files** (exported JSON) are not secret but reveal structure — fine to commit; don't put secrets in defaults.
- **Never destructive-fallback sensitive data** — losing a user's encrypted wallet data on a migration miss is both a data-loss and trust failure.
- **Version-gate security changes** — if a release tightens how sensitive data is stored, migrate old data into the new secure form (don't leave it in the old, weaker form).
- Client vs backend: migrations preserve **local** user data across versions; the backend remains authoritative but can't recover unsynced local data a bad migration destroys.

36.13 Performance and Resource Optimization

- **Migrations run at open (startup):** keep them fast; a slow table-recreate on a big table delays launch (Ch 88). Optimize SQL; add indexes **after** bulk copy.
- **Batch/transaction:** Room wraps each migration in a transaction — one commit, fast.
- **Incremental for huge data:** for very large transforms, consider migrating lazily or in the background to avoid a long startup stall.
- **Avoid unnecessary recreates:** use auto-migration/ `ALTER ADD COLUMN` for additive changes (no copy) instead of recreating.
- **Profiling:** migration duration per version (telemetry), startup time around releases, DB size.

36.14 Testing Strategy

`MigrationTestHelper` is **mandatory** — it creates an old-version DB from the exported schema, inserts data, migrates, and validates schema + data.

```
@RunWith(AndroidJUnit4::class)
class MigrationTest {
    private val TEST_DB = "migration-test"
    @get:Rule val helper = MigrationTestHelper(
        InstrumentationRegistry.getInstrumentation(),
        StreamCartDatabase::class.java,            // uses exported schemas/
    )

    @Test fun migrate1To2_addsDiscount_preservesData() {
        helper.createDatabase(TEST_DB, 1).apply {
            execSQL("INSERT INTO products (id, title, price) VALUES ('1','Shoe','12.99')")
            close()
        }
        val db = helper.runMigrationsAndValidate(TEST_DB, 2, true, MIGRATION_1_2)
        db.query("SELECT discount FROM products WHERE id='1'").use {
            it.moveToFirst(); assertEquals(0, it.getInt(0))        // new col default
        }
    }

    @Test fun migrate3To4_typeChange_transformsAndPreserves() {
        helper.createDatabase(TEST_DB, 3).apply {
            execSQL("INSERT INTO products (id, title, price, discount) VALUES
('1','Shoe','12.99',0)")
            close()
        }
        val db = helper.runMigrationsAndValidate(TEST_DB, 4, true, MIGRATION_3_4_TYPE_CHANGE)
        db.query("SELECT price_cents FROM products WHERE id='1'").use {
            it.moveToFirst(); assertEquals(1299, it.getInt(0))     // "12.99" -> 1299, data
preserved
        }
    }

    @Test fun migrateAll_1To5_endToEnd() {
        helper.createDatabase(TEST_DB, 1).close()
        helper.runMigrationsAndValidate(TEST_DB, 5, true,
            MIGRATION_3_4_TYPE_CHANGE, MIGRATION_4_5_FTS)          // + autos
    }
}
```

What to test: each migration (schema validates via exported JSON + **data preserved/ transformed**), full-chain 1→N, unsynced outbox survival, encrypted-DB migration (Ch 24), and a test that fails if someone adds destructive fallback. Run these in CI.

36.15 Production Observability

- **Metrics:** migration success rate per version (should be ~100%), crash-on-open rate around releases, migration duration, startup-time delta per release, DB size.
- **Logs:** `migration{from, to, ms, result}` (no row data, Ch 28), DB-open errors.
- **Debugging:** crash spike after release → missing/failing migration (add/fix + test); data-loss reports → destructive fallback or lossy transform; slow startup → heavy migration on big tables; corrupt data → bad transform SQL (test with data).
- **Alerting:** crash-on-open or migration-failure spike post-release → emergency (hotfix the migration); never "fix" with destructive fallback.
- **Pre-release gate:** CI runs `MigrationTestHelper` for all paths; block release on failure.

36.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Why migrate instead of recreate?* The local DB holds user data (incl. unsynced, Ch 31) — recreating loses it.
2. *What triggers a migration?* Incrementing `@Database(version)`.
3. *What happens with no migration path?* `IllegalStateException` on open (or data wiped if destructive fallback).
4. *`Migration(from,to)`?* A class whose `migrate(db)` runs SQL to evolve schema/data.
5. *Does Room chain migrations?* Yes — applies the sequence covering the user's installed → current version (or a direct path).

Auto vs manual

6. *Auto-migrations?* Room generates them from exported schemas for simple additive changes.
7. *`AutoMigrationSpec`?* Resolves ambiguous changes (rename/delete) via `@RenameColumn` / `@DeleteTable`.
8. *When manual?* Type/constraint changes, data transforms — auto can't express.
9. *Why export schemas?* Enables auto-migration, `MigrationTestHelper`, and code- review diffs; commit them.

Complex changes

10. *Change a column type in SQLite?* Can't ALTER — create new table, copy/transform, drop old, rename.
11. *Data transformation?* Do it in the copy SQL (e.g., `CAST`/compute).
12. *Recreate indexes/triggers?* Yes — after the table recreate.

Safety

13. *`fallbackToDestructiveMigration` danger?* Deletes the whole DB on a missing/ failed migration — forbidden for user data in release.
14. *When is it OK?* Purely disposable cache / debug; even then prefer cache versioning (Ch 34).
15. *Is a migration transactional?* Yes — fails and rolls back if the SQL errors.

Testing

16. *Why must migrations be tested?* The normal suite runs on a fresh DB; untested migrations crash/corrupt in production.
17. *How?* `MigrationTestHelper`: create old DB + data → migrate → validate schema + data.
18. *What to assert?* Schema matches exported JSON **and** data preserved/ transformed.

Relations

19. *vs API schema evolution (Ch 16)?* Storage schema vs network DTO — coordinate on model changes.
20. *vs cache versioning (Ch 34)?* DB migration preserves; cache versioning wipes stale cache shape — different data.

Security/perf

21. *Encrypted DB migration (Ch 24)?* Runs inside the SQLCipher DB; key at open; test it.
22. *Migrations slow startup?* Keep fast; index after copy; off-main open; incremental for huge data.

Production debugging

23. *Crash-on-open after release?* Missing/failing migration.
24. *Data-loss reports?* Destructive fallback or lossy transform.
25. *Corrupt data?* Bad transform SQL (test with data).
26. *Slow launch after update?* Heavy migration.

Design-change challenges

1. **Add a column safely.** Auto-migration (`ALTER ADD COLUMN DEFAULT`).
2. **Change price type.** Manual recreate + `CAST` transform; test data.
3. **Rename a column.** Auto-migration + `@RenameColumn` spec.
4. **Add FTS (Ch 35).** Migration creates the FTS virtual table + triggers.
5. **Crash-on-open shipped.** Hotfix the migration (never destructive fallback); add the missing path + test.

Mock walkthrough (abridged)

Clarify (what changed, data to preserve, unsynced?) → bump version → auto-migration for additive / manual recreate for type changes (transform in copy) → export + commit schemas → **test with `MigrationTestHelper`** (schema + data) → **never** destructive fallback → coordinate with Ch 16/34 → encrypted DB (Ch 24) → observability.

36.17 Practical Exercises

1. **Beginner.** Add a column via an auto-migration (or manual `ALTER`); export + commit the schema; test it preserves data. *Eval:* no crash; data intact.
2. **Intermediate.** Rename a column via `AutoMigrationSpec`; validate with `MigrationTestHelper`. *Eval:* renamed; data preserved.

3. **Advanced.** Change a column type via recreate+transform; test data is correctly converted. *Eval:* type changed; values transformed; no loss.
4. **Production debugging.** Fix a crash-on-open (missing migration) and remove a shipped `fallbackToDestructiveMigration`. *Eval:* migrates safely; no wipe.
5. **System design challenge.** Add an FTS table (Ch 35) and migrate an encrypted DB (Ch 24); test the full chain 1→N in CI. *Eval:* FTS searchable; encrypted migrate; all paths tested.

(Solutions withheld unless requested.)

36.18 Chapter Summary & Reference Sheet

Key decisions: evolve the local schema by **incrementing the version** and providing a **tested migration** that **preserves data**; use **auto-migrations** for additive changes (+ `AutoMigrationSpec` for renames/deletes) and **hand-written `Migration`s** for **type/constraint changes** (create-copy-transform-drop-rename); **`exportSchema = true`** and **commit** the JSON for auto-generation, review, and testing; **test every migration with `MigrationTestHelper`** (schema **and** data); **never `fallbackToDestructiveMigration()` in release** for user data (especially unsynced, Ch 31); **migrate encrypted DBs** (Ch 24); coordinate with API evolution (Ch 16) and cache versioning (Ch 34). **Provide a migration, never destroy.**

Concepts: schema version, `Migration` chain, auto-migrations + specs, complex table-recreate + data transform, schema export, `MigrationTestHelper`, destructive- fallback danger, encrypted-DB migration.

Common mistakes: bumping version without a migration (crash-on-open); `fallbackToDestructiveMigration` on user data (silent wipe); untested migrations (prod corruption); trying to ALTER a column type (recreate instead); not exporting/ committing schemas; losing unsynced outbox data; heavy migrations blocking startup.

Checklist:

- [ ] bump version
- [ ] auto-migration for additive / manual for complex
- [ ] data-preserving transform
- [ ] `exportSchema=true` + commit JSON
- [ ] `AutoMigrationSpec` for rename/delete
- [ ] test every migration (schema+data) in CI
- [ ] no destructive fallback (release)
- [ ] migrate encrypted DB (Ch 24)
- [ ] recreate indexes/triggers
- [ ] keep migrations fast.

Key APIs: `@Database(version, exportSchema=true, autoMigrations=[...])`, `@AutoMigration`, `AutoMigrationSpec` + `@RenameColumn` / `@RenameTable` / `@DeleteColumn` / `@DeleteTable`, `Migration(from,to)` / `migrate`, `SupportSQLiteDatabase.execSQL`, `MigrationTestHelper.createDatabase` / `runMigrationsAndValidate`, `room.schemaLocation`.

References:

- Room migrations: https://developer.android.com/training/data-storage/room/migrating-db-versions

- Auto-migrations: https://developer.android.com/training/data-storage/room/migrating-db-versions#automated
- Testing migrations: https://developer.android.com/training/data-storage/room/migrating-db-versions#test
- (composes) Ch 3 Room, Ch 16 API Evolution, Ch 24 Encryption, Ch 34 Cache Versioning, Ch 35 FTS

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; version/auto/manual/complex-recreate/FTS migrations + export + test; concurrency (transactional, startup); failure (crash-on-open, data loss); 5 levels; auto-vs- manual and destructive-fallback trade-offs; security (encrypted migrate, version-gate); perf (fast migrations); mandatory `MigrationTestHelper` tests; observability (migration success rate); 26 Q&A + 5 challenges. ✅
- **Expand later:** encrypted-DB specifics (Ch 24), FTS triggers (Ch 35), startup (Ch 88), coordinated multi-module schema governance.
- **Interview coverage:** exceeds ≥25 bar.